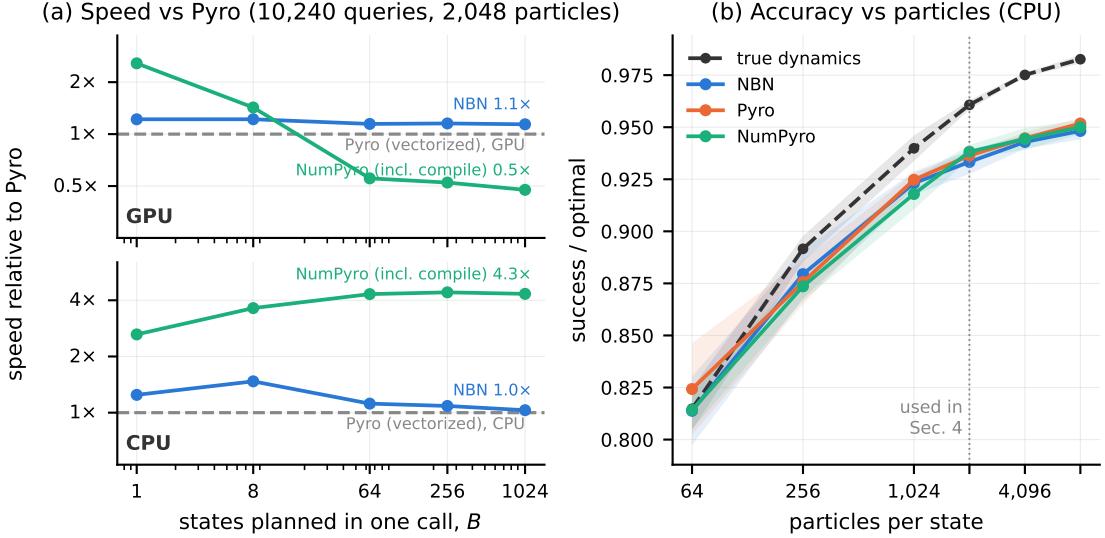
(a) Speed vs Pyro (10,240 queries, 2,048 particles)
speed relative to Pyro
2×
1×
0.5×
NBN 1.1×
Pyro (vectorized), GPU
NumPyro (incl. compile) 0.5×
GPU
NumPyro (incl. compile) 4.3×
4×
2×
1×
NBN 1.0×
Pyro (vectorized), CPU
CPU
1
8
64
256
1024
states planned in one call, B
(b) Accuracy vs particles (CPU)
success / optimal
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.800
true dynamics
NBN
Pyro
NumPyro
used in
Sec. 4
64
256
1,024
4,096
particles per state